

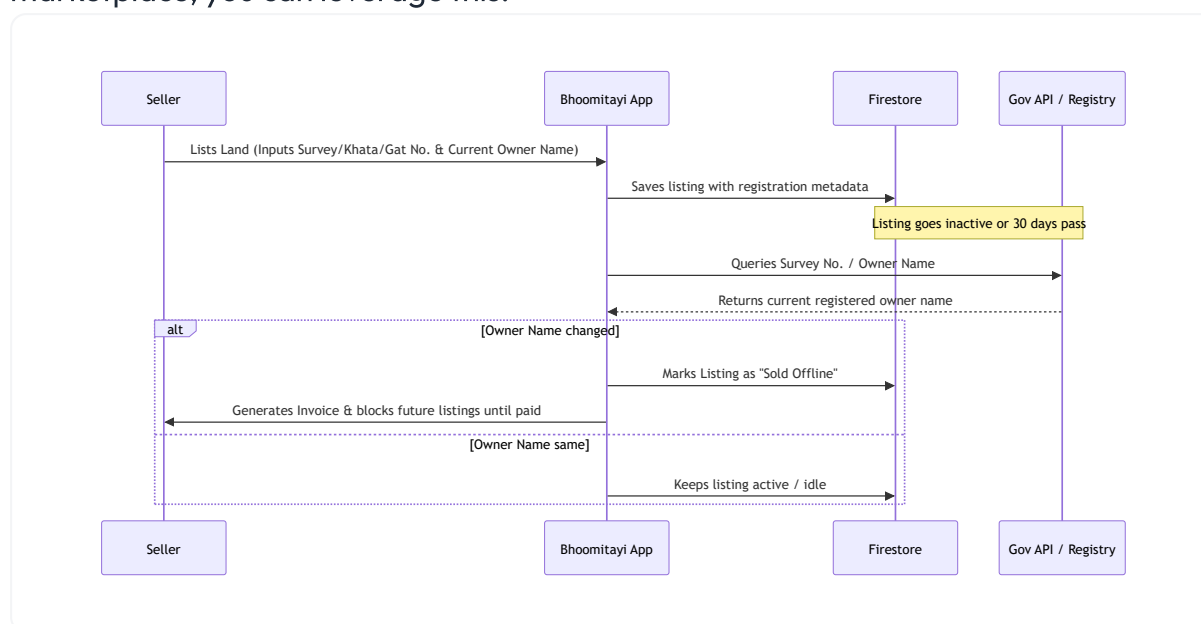
Deep Implementation Plan: Tracking Offline Property Sales for Bhoomitayi

Since sellers refuse to pay upfront or subscribe, and will only pay after a successful sale, your primary challenge is **discovering when a sale actually happens offline** so you can invoice them.

This plan details **four deep, practical systems** you can build into Bhoomitayi's Next.js and Firebase architecture to track, verify, and enforce post-sale payments.

System 1: Government Land Registry Check (The Foolproof Technical Method)

In India, land cannot be sold privately without official registration at the Sub-Registrar's office. This generates a public record. Since Bhoomitayi is a land/property marketplace, you can leverage this.



How to implement:

Mandatory Survey / Khata / Gat Numbers:

When a seller lists land, make the **Survey Number, Village, Taluk/Tehsil**, and **District** mandatory fields. (Explain to the user: *"This is required to verify the authenticity of your listing for buyers"*—which buyers love anyway).

Record the **Current Owner's Name** as it appears on the land records (e.g., Pahani/RTC in Karnataka, 7/12 extract in Maharashtra, Jamabandi in Punjab/Haryana).

Automated Registry Verification Service:

Integrate a third-party Land Record API provider (like Zoop.one, Land Registry APIs, or writing a custom parser/scrapper for state land records websites like Kaveri/Bhoomi, Mahabhulekh, AnyROR).

The Audit Job: Set up a Firebase Scheduled Cloud Function to run every 15 to 30 days for listings that are either:
Marked "deleted" or "inactive" by the user.

Idle with no updates for over 45 days.

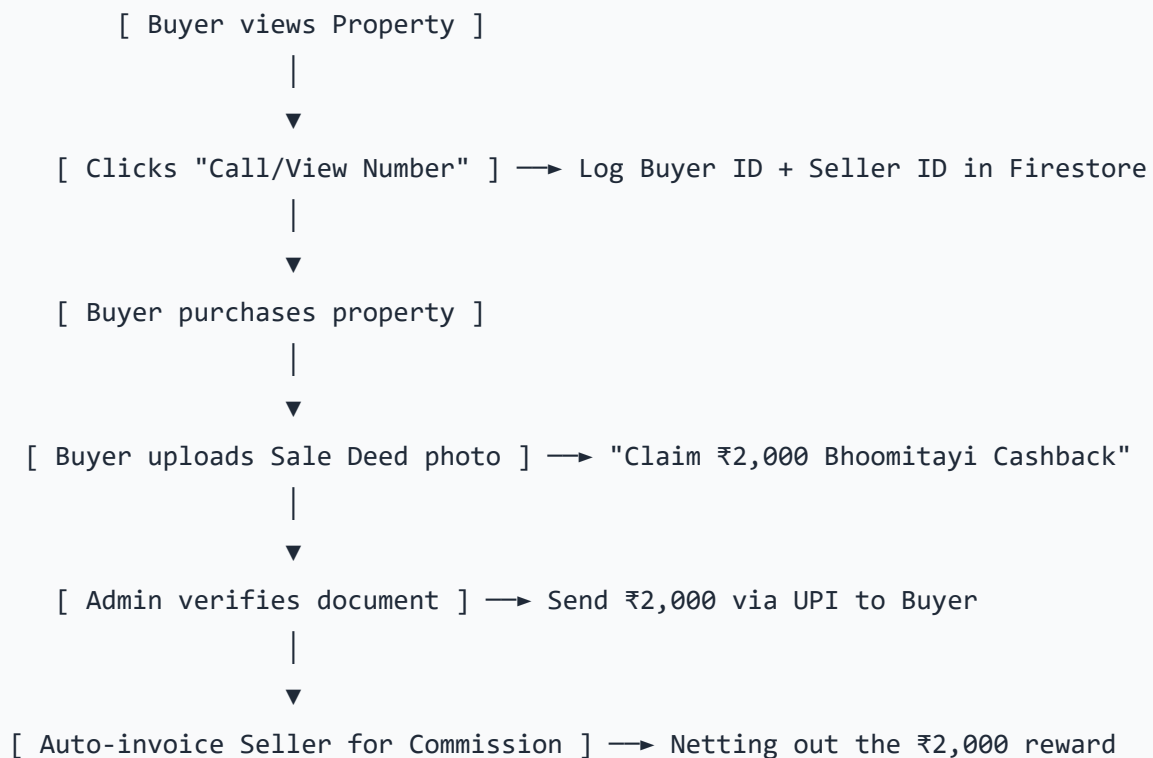
The Cloud Function queries the state land record database using the Survey Number.

The Check: Compare the current registered owner's name with the owner's name logged when the listing was created.

The Result: If the registered owner name has changed, **the land has officially been sold**. You now have legally verifiable proof of sale.

System 2: Buyer Verification with UPI Cashbacks (The Social Proof Method)

Buyers have no incentive to hide the sale because they aren't the ones paying the commission. You can incentivize them to report the transaction.



How to implement:

The Lead Capture:

When a buyer clicks "Call Seller" or "Chat", save a record in the `leads` collection in Firestore:

```
interface Lead {
  leadId: string;
  listingId: string;
  sellerId: string;
  buyerId: string;
  buyerPhone: string;
  timestamp: Date;
  status: 'initiated' | 'verified_purchase';
}
```

The Cashback Offer:

Display a persistent banner for buyers: *"Bought this property? Upload your Sale Agreement / Deed or Registration Token and get ₹1,000 - ₹2,000 instant cashback via UPI + Free Property Mutation (Khata Transfer) guidance."*

The Payout and Invoice Ledger:

When a buyer uploads proof, your admin dashboard verifies it.

Pay the buyer their cashback immediately (you can automate this via Cashfree Payouts API).

Automatically generate a Cashfree invoice for the seller.

If the seller refuses to pay, you have the buyer's uploaded document as absolute proof of sale to blacklist the seller, block their phone number/IP, and take legal recourse if your terms of service outline a contract.

System 3: Automated WhatsApp & Call Audit Bot (The Conversational Method)

Real estate transactions are slow and conversational. By automating the communication loop, you can extract sale information.

How to implement:

WhatsApp Cloud API Integration:

Use an official WhatsApp BSP (like Gupshup, Wati, or Aisensy) integrated with your Next.js backend.

The Verification Sequence:

If a seller marks a listing as "hidden", "deleted", or "inactive", trigger a WhatsApp interactive message flow to the **Buyer** and **Seller**:

To the Seller: *"Hi [Name], we noticed you paused your land listing in Mandya. Did you find a buyer?"*

Button 1: *"Yes, sold through Bhoomitayi"* \$

→ Auto-opens Cashfree payment link. * Button 2: *"Yes, sold elsewhere"* \$

→ Ask for buyer details to confirm. * Button 3: *"Not sold yet, just pausing"* \$

→ Snoozes for 14 days. * **To the Buyers who called this seller:** *"Hi [Name], did you purchase the land in Mandya from [Seller Name]?"* * Button 1: *"Yes, bought it"*

\$ → Triggers the Cashback form upload. * Button 2: *"No, still looking"* \$

→ Offer them other listings. 3. **Natural Language Processing (NLP)**

detection:

If they reply with text like "sold it last week" or "agreement done", use a basic AI parser to flag the listing status as `Sold Offline` in your admin panel.

System 4: Ground-Level Micro-Brokers ("Bhoomi Mitras")

In rural and semi-urban India, local brokers know exactly who bought what land. You can crowdsource verification.

How to implement:

Recruit Local Informants:

Allow local small brokers or village youth to sign up as "Bhoomi Mitras" (verification agents) on the app.

Audit Tasks:

When a high-value listing (e.g., > ₹10 Lakhs) goes inactive or is deleted without payment:

Generate a verification task in the area: *"Verify if Survey No. 45 in Village X is sold."*

Assign this to a nearby Bhoomi Mitra.

The Mitra goes to the village, asks around, and confirms if the land changed hands.

Incentive: If they prove it was sold, they get a flat payout (e.g., ₹500), and you charge the seller the success fee.

Comparison of Methods

Method	Technical Effort	Reliability	Setup Cost	Actionability
1. Gov Registry Check	High (APIs/Scrapers)	100% (Foolproof)	Low (per-lookup cost)	Best for agricultural land/plots
2. Buyer UPI Cashback	Medium (UPI Payouts)	High	Medium (Cashback fund)	Best for immediate user adoption
3. WhatsApp Bot	Low (Wati/Gupshup)	Medium	Very Low	Best for filtering active listings
4. Ground Brokers	Medium (Task flow)	High (Local Intel)	Commission-based	Best for high-value rural deals

Recommendation & Immediate Next Steps

If you want to implement this without changing code today, write down these product requirements:

Database Schema updates: Prepare to collect **Survey Numbers** during listing submission.

Lead Tracking: Prepare to capture `userId` and `timestamp` whenever a contact details button is clicked.

Cashback Promotion UI: Draft a design where buyers are prompted to submit sale agreements for cash rewards.

Draft Terms of Service: Add a clause stating that by listing a property, the seller agrees to let Bhoomitayi verify the registry records for the property, and that deleting the listing without notification is considered a breach if a sale occurred.